

Data Cleanser — Theory Concepts

Missing Value Imputation & Outlier Detection/Removal — Definitions & Rationale

1. Missing Data

Missing data occurs when no value is stored for a variable in an observation. It is commonly classified into three mechanisms:

- **MCAR (Missing Completely At Random):** The probability of a value being missing is unrelated to any observed or unobserved data. (Used to generate this project's synthetic missingness.)
- **MAR (Missing At Random):** The probability of missingness depends on other observed variables, but not on the missing value itself.
- **MNAR (Missing Not At Random):** The probability of missingness depends on the missing value itself (e.g., very sick patients skipping a checkup).

2. Imputation Techniques

Simple Imputer (Numerical) — Mean / Median

Replaces missing numeric values with the column mean or median. Mean imputation is sensitive to outliers and skew; median imputation is more robust. Both reduce the variance of the column, since a single repeated value replaces many distinct values.

Simple Imputer (Categorical) / Most Frequent Imputation

Replaces missing categorical values with the mode (most frequent category). Simple and fast, but can amplify class imbalance in the categorical variable if used heavily.

Missing Indicator + Random Sample Imputation

Two-part strategy: (1) create a binary flag column recording *whether* a value was originally missing (preserves that signal for the model), and (2) fill gaps by drawing a random sample from the column's own observed distribution, which preserves the original variance and shape far better than mean/median imputation.

KNN Imputer

A multivariate method: for each missing value, find the k most similar rows (nearest neighbours) based on the other available features, then impute using their (distance-weighted) average. Captures relationships between features rather than treating each column independently.

MICE Algorithm (Multiple Imputation by Chained Equations)

An iterative, multivariate method: each column with missing values is modeled as a function of all other columns (e.g. via regression), cycling column-by-column across several iterations until the imputed values converge/stabilize. Scikit-learn's *IterativeImputer* implements this chained-equations approach and is used in this project with a Random Forest regressor as the per-column estimator.

3. Outlier Detection & Treatment

Z-score Method

Standardizes a column to a mean of 0 and standard deviation of 1, then flags any value whose absolute z-score exceeds a threshold (commonly 3). Assumes an approximately normal distribution; sensitive to the very outliers it is trying to detect (the mean and standard deviation themselves shift under extreme values).

IQR (Interquartile Range) Method

Computes Q1 (25th percentile) and Q3 (75th percentile); the IQR = Q3 – Q1. Any value below Q1 – 1.5×IQR or above Q3 + 1.5×IQR is flagged as an outlier. Distribution-free (does not assume normality), and is the basis for the whiskers on a boxplot.

Percentile Method

Defines fixed cut-off percentiles (commonly the 1st and 99th) and flags/caps any value outside that range. Simple and directly interpretable, but the choice of percentile threshold is somewhat arbitrary and dataset-dependent.

Winsorization

Rather than deleting outlier rows, Winsorization *caps* extreme values at a chosen percentile boundary (e.g., anything above the 99th percentile is set equal to the 99th percentile value). This reduces the distortion outliers cause to means/variances while retaining every observation (and all of that row's other, valid measurements) in the dataset — important when the dataset is not very large, as is the case here.

4. Why Treat Outliers by Capping Rather Than Deleting?

Deleting every row flagged by an outlier test discards potentially valuable information (e.g. a genuinely very sick patient's other, valid measurements) and shrinks the dataset available for model training. Winsorization keeps the sample size intact while still preventing extreme values from disproportionately influencing distance-based algorithms (like KNN) or gradient-based algorithms (sensitive to feature scale/outliers).

5. Summary Comparison

Technique	Type	Handles	Preserves Variance?
Simple Imputer (mean/median)	Univariate	Missing values	No — shrinks it
Most Frequent Imputation	Univariate	Missing categorical	Reduces category diversity
Missing Indicator + Random Sample	Univariate + flag	Missing values	Yes
KNN Imputer	Multivariate	Missing values	Yes
MICE (IterativeImputer)	Multivariate, iterative	Missing values	Yes — best
Z-score	Univariate	Outlier detection	N/A (detection only)
IQR	Univariate	Outlier detection	N/A (detection only)
Percentile	Univariate	Outlier detection	N/A (detection only)
Winsorization	Univariate	Outlier treatment	Reduces extreme variance, keeps rows